

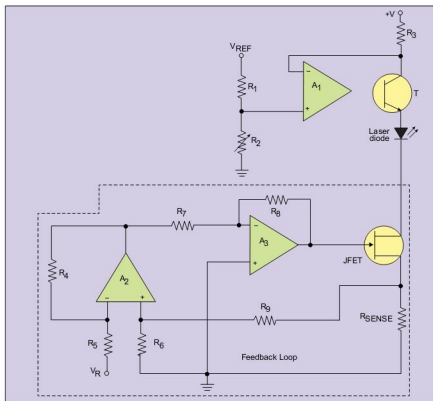


Fig. 6: Laser-diode precise current control



Fig. 7: Precision constant-current laser diode driver for OEM applications

One such circuit is shown in Fig. 6. The basic circuit topology is similar to the one used in driver circuits described in the previous pages. The only change is inclusion of a junction FET (JFET) connected in series with the sense resistor and wired as a voltage variable resistor. A small variation in drive current is compensated by an appropriate variation in the drain-source resistance of the JFET.

Initially, at the nominal value of drive current, circuit parameters are so adjusted as to ensure that JFET with the feedback loop closed gets a negative gate voltage to keep it nearly in the middle of its VVR characteristics. This is done to fully exploit the voltage-

are commercially available today from a host of manufacturers for original equipment manufacturer (OEM) applications. A representative photograph is shown in Fig. 7.

Protection features, such as slow start, immunity to fast transients and overcurrent limit, as outlined earlier, are essential features for every laser diode driver circuit to have. Fig. 8 shows a constant-current laser diode drive circuit with modulation capability and in-built above-mentioned protection features.

Let us first talk about protection features provided by the drive circuit. The Schmitt comparator at the input provides a delay of the order of a few tens of milliseconds after the switch-on to offer protection against switch-on transients. The time delay is decided by the R_1C_1 time constant. R_2 and C_2 provide the desired slow or soft start and decay during switch on and switch off. R_2C_2 time constant is of the order of one to two seconds. R_{11} and C_3 provide

additional protection against transients; D_1 protects the laser diode against reverse voltages.

The constant value of drive current is decided by the voltage present at the noninverting input of op-amp A_4 , which equals the voltage present at the output of op-amp A_2 , provided $R_8 = R_9$. This is further equal to sum of voltage levels

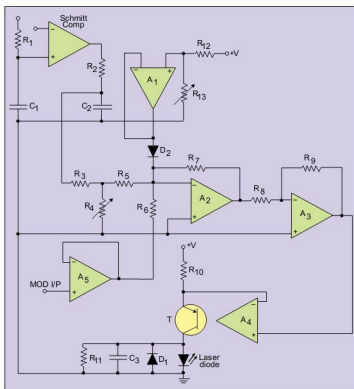


Fig. 8: Constant-current drive circuit with protection features

dependent resistance range. Precision constant-current laser diode drivers

due to modulation input and reference voltage V_R after potential divider